Employee Stock Purchase Plan

Our employee stock purchase plan ("ESPP") was offered to substantially all employees beginning in August 2018 and permits eligible employees to purchase shares of our common stock through payroll deductions of up to 10% of their eligible compensation, subject to certain limitations. The purchase price of the shares under the ESPP equals 85% of the lower of the fair market value of our common stock on either the first or last day of each six-month offering period. Compensation expense is calculated as of the beginning of the offering period as the fair value of the employees' purchase rights utilizing the Black-Scholes option valuation model and is recognized over the offering period. Assumptions used in the Black-Scholes option valuation model were as follows:

For the year ended	2019	2018
Weighted-average grant-date fair value per share	$ 11.60	$ 14.55
Average expected life in years	0.5	0.5
Weighted-average expected volatility	45.0%	43.0%
Weighted-average risk-free interest rate	2.2%	2.2%
Expected dividend yield	0.0%	0.0%

Employees purchased 3 million shares of common stock for $95 million in 2019 under the ESPP.

Stock-based Compensation Expense

For the year ended	2019	2018	2017
Stock-based compensation expense by caption			
Cost of goods sold	$ 102	$ 83	$ 88
Selling, general, and administrative	73	61	75
Research and development	68	54	52
	$ 243	$ 198	$ 215
Stock-based compensation expense by type of award			
Restricted stock awards	$ 178	$ 140	$ 144
Stock options	33	55	71
ESPP	32	3	—
	$ 243	$ 198	$ 215

The income tax benefit related to share-based compensation was $66 million, $158 million and $97 million for 2019, 2018 and 2017, respectively. The income tax benefits related to share-based compensation for the periods presented prior to the second quarter of 2018 were offset by an increase in the U.S. valuation allowance. Stock-based compensation expense of $30 million and $19 million was capitalized and remained in inventory as of August 29, 2019 and August 30, 2018, respectively. As of August 29, 2019, $439 million of total unrecognized compensation costs for unvested awards, before the effect of any future forfeitures, was expected to be recognized through the fourth quarter of 2023, resulting in a weighted-average period of 1.3 years.

Employee Benefit Plans

We have employee retirement plans at our U.S. and international sites. Details of the more significant plans are discussed as follows:

Employee Savings Plan for U.S. Employees

We have a 401(k) retirement plan under which U.S. employees may contribute up to 75% of their eligible pay, subject to Internal Revenue Service annual contribution limits, to various savings alternatives, none of which include direct investment in our stock. We match in cash eligible contributions from employees up to 5% of the employee's annual eligible earnings. Contribution expense for the 401(k) plan was $67 million, $61 million, and $52 million in 2019, 2018, and 2017, respectively.